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1. Edge computing & Fog computing

→ Edge computing is distributed computing paradign where data processing occurs to the data source rather than sending all data to a centralized could server.

→ In Traditional colud computing, devices send data to a ~~le~~ remark data center for processing However, this creaks latency, bandwidth usage and slower response times. Edge computing ~~slower~~ solves this problem by processing data locally on devices or never by serves.

→ charac teristics of Edge computing.

→ Processing happens near the user or device.

→ low catentey and faster response time.

→ Reduces band width consymption.

→ Suppots real time applications.

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→ Advantage :-

1. Faster data Processing and Respons time.
2. Reduced Network congetion.

→ Applications
→ Autonous vechicles
→ smart cities
→ Healtcare monitoring devices
→ Fog computing
→ Fog computing Extends cloud computing by Placing computing resources between edge devices and the cloud.

→ It acts as an inter medidate layer that Processes and filters data before sending it to the cloud.

→ Fog computing was introduced by cisco to improve efficiency in IO system.

→ characteristics of Fog computing.

→ Distributed computing architec ture.

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→ Techniques used
→ virtualization
→ Efficent Data centers
→ Renewable Energy.
→ Dynamic Resource Allocation.

→ Examples of Green cloud Providers.

→ major cloud compnies focusing on sustainability include.

→ Amzon web services

→ google clould

→ Microsoft Azure.

→ Benfits -

- Reduced operational cost
- lower environmental Impact
- Improved resource utilization
- sustainable technology development.

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→ Architecture work flow
→ collect data
→ Store data in Aws sagemaker
→ Evaluate model Performance.
→ Deploy model as an endpoint.

→ Applications.-

→ Recommedation system
→ fraud detection
→ Image recognition
→ Benefits
→ fully manged infrastruce
→ scalable training and deployment.

4. Green Cloud computing and susloinability.

→ Green cloud computing fouses on designing and using cloud computing system is a way that reduce aniironmtal imput and energy consumption.

→ objetives of Green cloud computing.

→ Reduce Power consumption.

→ lower carbon emissions.

→ improve enegy efficiency.

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→ Quantum Hundwere is Expensive.

→ Error raks are High.

3. AI / ML in cloud - AWS sage maker.

→ AI and machine learing in cloud compting allow developers to loyild, train, and deploy machine learning models using cloud infrastructur.

→ one of the most Popular cloud service for ML is Amzon Sagemaker, Provided by Amzone web services.

→ what is AWS sagemaker

→ Awa sage makes is fully manged machine learning servies that keys

developers and data scientists build train and deploy ml models quickly and efficently.

→ key featurs of AWS sage maker.

1. Data Perpration
2. Model training
3. Model deployment
4. Automtic model Taming

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→ superposition -

superposition allows quantum computers to process multiple possibilites simaltaneosly.

→ Entanglement -

Entorglent allows Qubits to be correlated with each other even at long distances, enabling faster computations.

→ Quantum gates -

Quantum takes manipulate Qubits similar to logic gates in classical computing.

→ Quantum cloud Platforms.

IBM Quantum Experience
google Quantum AI

→ Applicatinon of Quantum cloud computing.

→ cryplography and security

→ Drug discovery and molecular simulation.

→ challenges

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→ Data Processing occurs at local network nodes.

→ Supports real time analytics.

→ Both edge and fog computing help reduce latency and improve efficiency in IoT and real-time system by bringing computation closer to data source.

2. Quantum cloud computing.

→ Quantum cloud computing refers to Providing Quantum computing resources through cloud Platforms so users can run Quantum algorithums without owing Questum hardware.

→ Quantum computers use Qubits instead of classical bits unlike bits that represent 0 or 1, qubuits can Exist in multiple states, simyltaneously due to quantum mechinics.

→ key concepsts :-

→ Qubit

A Qubit is basic unit of Quantum information.